2021 HW1

MOS Theory, Multiplexers, and Static CMOS

EE3032 / Introduction to VLSI Design

Provenance. The original assignment and the supplied answers by Lin Yan-Yu were read first. The results below were independently derived, with the useful reference choices retained and improvements identified. Source details and hashes are in the year-level `SOURCES.md`.

Notation. + is OR; juxtaposition is AND. Every pMOS body is tied to V_{DD} and every nMOS body to ground. Body terminals are omitted in the diagrams. Only true input literals are used in the CMOS gates below.

1. Two identical series devices per inverter branch

(a) The rail-side devices M_1 and M_4 cannot carry nonzero DC current in saturation. For the nMOS stack let the internal node be u and $U = V_{\mathrm{in}} - V_{tn} > 0$. Saturation of the lower M_1 would require $u \geq U$. But then the upper device has $V_{GS,2} = V_{\mathrm{in}} - u \leq V_{tn}$ and is cut off. Thus the assumed positive stack current is impossible. The pMOS argument is identical with voltages measured downward from V_{DD}: saturation of rail-side M_4 would cut off M_3. At cutoff or the zero-current boundary, the usual region inequalities may coincide; the statement concerns conducting operation. **(b) Both equivalent aspect ratios are one half of the original:**

$$\boxed{(W/L)_{N_1} = (W/L)_{N_2} = \tfrac{1}{2}(W/L).}$$

Here N_1 is nMOS and N_2 is pMOS. This equality uses the ideal long-channel square-law model, no body effect, and no channel-length modulation; it does not assert exact BSIM short-channel equivalence. Let $k_n = \mu_n C_{\mathrm{ox}}(W/L)$. If the upper nMOS is saturated,

$$k_n(Uu - u^2/2) = \frac{k_n}{2}(U-u)^2, \quad u = U\left(1 - \frac{1}{\sqrt{2}}\right), \quad I = \frac{k_n U^2}{4}.$$

A single saturated nMOS with $k_{\mathrm{eq}} = k_n/2$ has the same current. If both devices are linear and $v = V_{\mathrm{out}} < U$, adding their equal-current equations gives

$$\frac{2I}{k_n} = Uv - \frac{v^2}{2}, \qquad I = \frac{k_n}{2}\left(Uv - \frac{v^2}{2}\right).$$

Thus the half-aspect-ratio device also matches the linear part of the branch characteristic; the pMOS derivation is symmetric. Halve W at fixed L, or double L in this ideal model. **Reference comparison.** The supplied answer identifies M_1, M_4 and the factor $1/2$ correctly. The present proof also covers the linear region. `verify_logic.py` checks 5,050 normalized operating points; the maximum current-equation error is below 10^{-12}.

2. Exactly two of three inputs are one

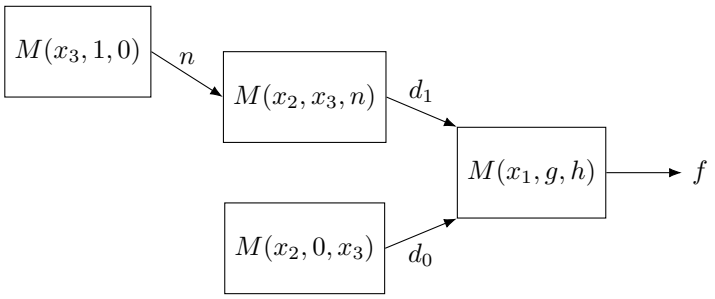
The requested function is

$$f = \bar{x}_1 x_2 x_3 + x_1 \bar{x}_2 x_3 + x_1 x_2 \bar{x}_3 = \bar{x}_1 (x_2 x_3) + x_1 (x_2 \oplus x_3).$$

Write $M(s, d_0, d_1) = \bar{s} d_0 + s d_1$. An implementation using four 2:1 multiplexers and constants only is

$$n = M(x_3, 1, 0), \quad g = M(x_2, 0, x_3), \quad h = M(x_2, x_3, n), \quad \boxed{f = M(x_1, g, h).}$$

The first mux makes $\bar{x}_3$ without an inverter; $g = x_2 x_3$ and $h = x_2 \oplus x_3$.



x_1	x_2	x_3	f
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

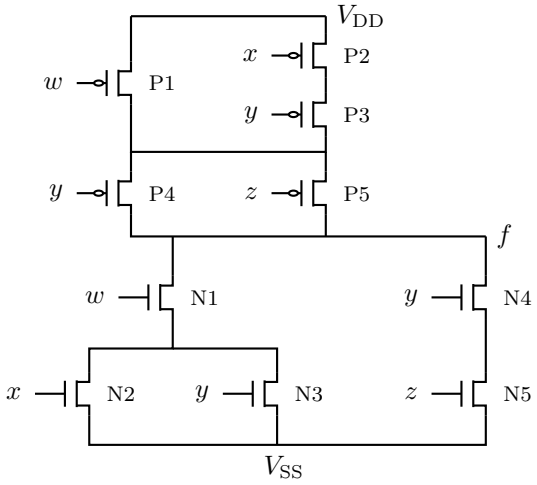
Reference comparison. The supplied full Shannon tree has seven muxes, including constant-only leaves. Its truth table is correct. The factored four-mux circuit saves three muxes while still using muxes exclusively. Exhaustive evaluation of all eight inputs confirms equality.

3(a). Complemented AOI function

Read the overbar in the prompt carefully: it spans the entire sum.

$$\boxed{f(w,x,y,z) = \overline{w(x+y)+yz}.}$$

The pull-down network (PDN) implements $G = w(x+y)+yz$: n_w in series with $(n_x \parallel n_y)$, in parallel with (n_y series n_z). The pull-up network (PUN) is the dual: $(p_w \parallel (p_x \text{ series } p_y))$ in series with $(p_y \parallel p_z)$.



There are five nMOS and five pMOS devices: **10 transistors total**. Two devices are separately controlled by y in each network; they are distinct transistors.

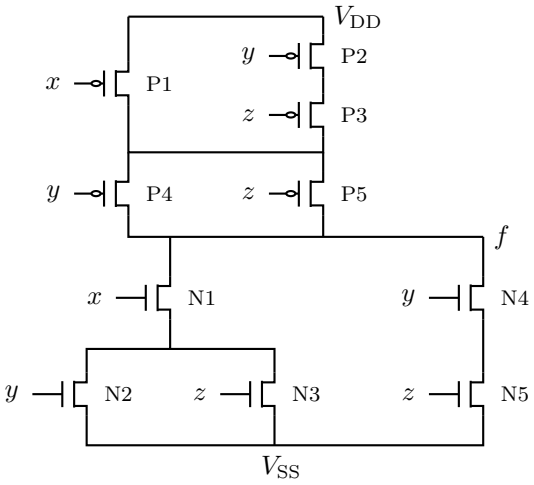
$$PUN = [\bar{w} + \bar{x}\bar{y}](\bar{y} + \bar{z}) = \overline{w(x+y)+yz} = f.$$

Every input has exactly one conducting rail network in ideal switch logic, so there is no static VDD-to-ground path. No complemented external input is needed. **Reference comparison.** The supplied diagram uses the same correct 10-transistor topology. This solution retains it, clarifies the overbar, and checks all 16 input combinations.

3(b). Complement of a three-input majority

$$f = \bar{x}\bar{y} + \bar{x}\bar{z} + \bar{y}\bar{z} = \overline{xy + xz + yz} = \boxed{\overline{x(y+z) + yz}}.$$

Use the same network structure as 3(a), now with the first branch n_x in series with $(n_y \parallel n_z)$ and the second branch n_y in series with n_z.



Gate labels in this diagram. P4/N4 are controlled by y and P5/N5 by z; repeated labels denote independent devices. The dual PUN is $(p_x \parallel (p_y \text{ series } p_z))$ in series with $(p_y \parallel p_z)$. It uses true literals only and needs **10 transistors**.

x	y	z	f
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

Reference comparison. The supplied answer implements the equivalent PDN $(x+y)(x+z)(y+z) = xy + xz + yz$, using six nMOS and six pMOS. Factoring the majority expression saves two transistors. All eight combinations pass the independent PDN/PUN complement check.